CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-501: Introduction to DBMS

Session: December 2013 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Define Database and Database Management System.

Definition: A Database is an organized collection of logically related data. A DBMS is the software interface that controls data storage, retrieval, and access permissions.

Q2 (3M) What is Data Redundancy?

Definition: Unnecessary duplication of identical data items across multiple database files, causing wasted storage and risk of inconsistency.

Q3 (3M) Define Primary Key.

Definition: A unique attribute chosen to identify tuples in a relation. Cannot accept duplicate values or NULL.

Q4 (3M) What is Relational Data Model?

Definition: A data model representing data in terms of mathematical relations (2D tables) composed of rows (tuples) and columns (attributes).

Q5 (3M) What is a Transaction?

Definition: A logical unit of database processing that includes one or more database access operations, executed as an indivisible atomic action.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain the Three-Level ANSI/SPARC Database Architecture.

Architecture: External, Conceptual, and Internal levels providing logical and physical data independence.

Q7 (7.5M) Explain Hashing and all collision resolution techniques in detail.

Collision Resolution: Detailed explanation of Linear Probing, Quadratic Probing, Chaining with overflow buckets, and Extendible Hashing.

Q8 (7.5M) What do you understand by Transaction System? Explain ACID properties.

Transaction Management: Atomicity, Consistency, Isolation, and Durability guaranteed through concurrency control and recovery.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) What is Normalization? Explain in detail 1st, 2nd, and 3rd Normal Form with importance.

Complete Normalization: 1NF (atomic values), 2NF (elimination of partial dependencies), and 3NF (elimination of transitive dependencies) with complete table decomposition walkthroughs.

Q10 (15M) **Explain Relational Algebra Operations with expressions and examples.**

Relational Algebra: Complete coverage of Selection, Projection, Cartesian Product, Union, Difference, and Natural Join.

Q11 (15M) **Explain ER Diagram concepts: Entity Sets, Weak Entities, Attributes, Relationships, and Cardinality. Draw ER diagram for Banking System.**

ER Model: Detailed conceptual design for Customer, Account, Branch, and Loan entities with conversion to relational tables.

Q12 (15M) **Explain File Organization Techniques: ISAM, B-Tree, and B+ Tree with diagrams.**

File Organization: Detailed structural differences and access speeds between Indexed Sequential, B-Tree, and B+ Tree structures.

Q13 (15M) **Explain Concurrency Control Protocols: Two-Phase Locking (2PL) and Timestamp Ordering.**

Concurrency Control: Strict 2PL, Rigorous 2PL, Thomas Write Rule, and timestamp-based conflict resolution.

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SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Differentiate between Sequential and Random Access in files.

Differences: Sequential access reads records consecutively in order from the beginning of the file (O(N)). Random (direct) access jumps immediately to any record block by its calculated address or key index (O(1)).

Q2 (3M) What do you mean by Relational Algebra?

Definition: A procedural mathematical query language consisting of a collection of operations that take one or two relations as input and produce a new relation as output.

Q3 (3M) What are the various Relational Algebra operators?

Operators: Selection (\sigma), Projection (\pi), Union (\cup), Set Difference (-), Cartesian Product (\times), Rename (\rho), Join (\bowtie), and Division (\div).

Q4 (3M) Define Primary Key and Foreign Key.

Definitions: Primary Key uniquely identifies tuples in a table. Foreign Key references the primary key in another table, enforcing referential integrity.

Q5 (3M) What is Data Inconsistency?

Definition: Different duplicate copies of the same data item holding contradictory, out-of-sync values across multiple files.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain DBMS Architecture in detail.

Architecture: Complete ANSI/SPARC 3-schema architecture with internal, conceptual, and external levels.

Q7 (7.5M) Differentiate between Association and Aggregation with suitable examples.

Differences: Association represents a general structural relationship between entities (e.g., Student *enrolled in* Course). Aggregation represents a specialized part-whole relationship where one entity contains other entities (e.g., University *aggregates* Departments).

Q8 (7.5M) Explain B-Tree and B+ Tree File Organization.

Explanation: Self-balancing tree indexes where B+ Trees store all data pointers exclusively in linked leaf nodes, maximizing branching factor and range query performance.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) Given tables: DEPT(Deptno, Dname, Loc), PROJECT(Projno, Deptno, Projname). Write SQL queries for project allocations and department summaries.

SQL Solutions: Complete SQL join queries, aggregate counts per department, and filter criteria.

Q10 (15M) **What do you mean by Normalization? Why do we normalize? Explain 1NF, 2NF, 3NF, and BCNF.**

Comprehensive Normalization Guide: Full mathematical criteria, functional dependencies, partial and transitive dependency elimination, and BCNF decompositions.

Q11 (15M) **Explain Concurrency Control: Locking Protocols (2PL) and Serializability.**

Concurrency Control: In-depth coverage of Conflict Serializability, Precedence Graphs, Shared/Exclusive locks, and 2-Phase Locking.

Q12 (15M) **Draw an ER Diagram for a Hospital Management Enterprise.**

ER Model: Complete diagram for Doctors, Patients, Beds, and Medical Records with mapping to relational schema.

Q13 (15M) **Explain Log-Based Recovery and Checkpoint Mechanisms in DBMS.**

Database Recovery: Deferred update, immediate update, write-ahead logging (WAL), and checkpoint-based fast recovery.

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SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Define Database Management System (DBMS).

Definition: A DBMS is software designed to manage, store, query, and maintain structured data securely and reliably, guaranteeing data independence and concurrent multi-user access.

Q2 (3M) Differentiate between Schema and Instance.

Comparison: A Schema is the overall structural blueprint designed once, while an Instance is the actual operational collection of data tuples present at any given moment.

Q3 (3M) What is Data Independence?

Definition: The capacity to change the data schema at one architectural level without altering schemas at higher levels.

Q4 (3M) What do you understand by Hashing? Explain with example.

Definition: Direct access technique mapping search keys to physical bucket addresses via hash function h(k) = k \pmod m for O(1) lookups.

Q5 (3M) What are Nested SQL Queries? Explain with example.

Definition: A query enclosed inside another SQL query, passed via `WHERE` or `HAVING` clauses using operators like `IN`, `EXISTS`, `ALL`, `ANY`.

```
SELECT EmpName FROM Employee
WHERE Salary > (SELECT AVG(Salary) FROM Employee);
```

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain the Three-Level Architecture of DBMS and its primary objectives.

Architecture: View Level → Conceptual Level → Internal Storage Level. Primary objective is achieving program-data independence and providing customized views to multiple users.

Q7 (7.5M) Explain BCNF (Boyce-Codd Normal Form) and compare with 3NF.

Explanation: BCNF requires every determinant in a non-trivial functional dependency to be a super key. Stricter than 3NF, eliminating all redundancy based on functional dependencies.

Q8 (7.5M) Explain Concurrency Control and ACID properties.

Explanation: Mechanisms ensuring Atomicity, Consistency, Isolation, and Durability when multiple transactions access the database simultaneously.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain ER Modeling: Entities, Attributes, Relationships, and Cardinality Constraints. Draw complete ER diagram for College Administration.**

Full Analysis: Complete conceptual ER model with Student, Faculty, Department, and Course entity sets, relationships, and cardinalities.

Q10 (15M) **Define Normalization. Why do we need to normalize? Explain 1NF, 2NF, and 3NF with examples.**

Full Normalization Process: Comprehensive breakdown of insertion, deletion, and update anomalies, followed by step-by-step normalization through 1NF, 2NF, and 3NF.

Q11 (15M) **Explain Relational Algebra Operations with expressions and SQL equivalents.**

Operations: Selection (\sigma), Projection (\pi), Cartesian Product (\times), Union (\cup), Set Difference (-), and Natural Join (\bowtie).

Q12 (15M) **Explain Indexing and Hashing: Primary vs Secondary Index, Dense vs Sparse Index.**

Indexing Schemes: Clustered/Primary indexing on ordered files vs Non-clustered/Secondary indexing on unordered candidate keys, dense vs sparse index trade-offs.

Q13 (15M) **Explain Log-Based Recovery and Checkpoints.**

Recovery: Detailed explanation of Undo/Redo protocols, Write-Ahead Logging (WAL), system checkpoints, and crash restart recovery algorithms.

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SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Differentiate between Database System and File System.**

Differences: DBMS offers centralized control, eliminated data inconsistency, program-data independence, and built-in transaction recovery, which file systems lack.

Q2 (3M) **Explain Foreign Key with ON DELETE CASCADE.**

Explanation: A foreign key links a child record to a parent record. With `ON DELETE CASCADE`, deleting a parent row automatically cascades and deletes all corresponding child rows, maintaining referential integrity.

Q3 (3M) **What is a Weak Entity Set? How is it identified?**

Definition: An entity set lacking sufficient attributes to form a primary key. Identified using a *discriminator (partial key)* combined with the primary key of its identifying strong entity set.

Q4 (3M) **Differentiate between Natural Join and Theta Join.**

Differences: Theta Join (R \bowtie_\theta S) combines rows based on any arbitrary condition. Natural Join (R \bowtie S) automatically equates attributes having identical names and eliminates duplicate columns.

Q5 (3M) **Explain COMMIT and ROLLBACK commands in SQL.**

TCL Commands:

- `COMMIT` : Permanently saves all transaction modifications to disk.
- `ROLLBACK` : Reverts all uncommitted changes back to the start of the transaction or savepoint.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Explain BCNF (Boyce-Codd Normal Form) with example.**

Explanation: A relation is in BCNF if for every non-trivial functional dependency X \to Y, X is a Super Key. Eliminates anomalies arising from overlapping candidate keys.

Q7 (7.5M) **Compare Dense Index and Sparse Index in database indexing.**

Comparison:

- **Dense Index:** Contains an index record for *every single search-key value* in the data file. Faster search, larger index size.
- **Sparse Index:** Contains index records for only some search-key values (typically one per disk block). Consumes less space, but requires scanning within blocks.

Q8 (7.5M) **Explain Deadlock Detection and Prevention methods in DBMS.**

Methods: Deadlock prevention uses timestamp ordering (Wait-Die and Wound-Wait schemes). Deadlock detection periodically constructs a Wait-For Graph (WFG) and aborts a cycle-causing transaction.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Draw a detailed ER Diagram for a Banking Enterprise.**

Entities: `CUSTOMER`, `ACCOUNT`, `BRANCH`, `LOAN`.
Relationships: Customer deposits in Account, Customer borrows Loan, Branch maintains Accounts and Loans.

Q10 (15M) **Explain Relational Algebra operators: Selection, Projection, Cartesian Product, Join, and Division.**

Relational Algebra: Complete explanation of fundamental algebraic query operators with mathematical notations and practical examples.

Q11 (15M) **Explain Nested Subqueries and Correlated Subqueries in SQL with examples.**

Subqueries:

- **Nested Subquery:** Inner query executes once, and its result is used by outer query (e.g., `WHERE Salary > (SELECT AVG(Salary) FROM Emp)`).
- **Correlated Subquery:** Inner query references columns of the outer query and re-executes for each row evaluated by outer query.

Q12 (15M) **Explain ACID Properties and Serializability in transaction management.**

ACID & Serializability: In-depth coverage of Atomicity, Consistency, Isolation, Durability, conflict serializability, and conflict equivalence of schedules.

Q13 (15M) **Explain ISAM and B+ Tree File Organizations with comparative analysis.**

File Organization: Detailed architectural comparison between static ISAM hierarchy and dynamic self-balancing B+ Trees.

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SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Define Metadata and Data Dictionary.

Definition: Metadata is "data about data" describing database structures, constraints, and data types. A **Data Dictionary** is the repository that permanently stores this metadata.

Q2 (3M) Define Tuple, Relation, Attribute, and Domain.

Definitions:

- **Relation:** A two-dimensional table of records.
- **Tuple:** A single row in a relation representing an individual entity instance.
- **Attribute:** A named column describing a specific property.
- **Domain:** The set of allowable atomic values for an attribute.

Q3 (3M) What are Cardinality and Participation Constraints in ER Modeling?

Definitions:

- **Cardinality Ratio:** Specifies the maximum number of relationship instances that an entity can participate in (1:1, 1:N, M:N).
- **Participation Constraint:** Specifies whether existence of an entity depends on its being related to another entity: *Total* (double line) or *Partial* (single line).

Q4 (3M) Differentiate between DDL and DML.

DDL: Defines database schemas and tables (`CREATE` , `ALTER` , `DROP`).
DML: Manipulates and retrieves data instances (`INSERT` , `UPDATE` , `DELETE` , `SELECT`).

Q5 (3M) What is a View in SQL?

Definition: A **View** is a virtual table whose contents are defined by an underlying SQL query, providing security and abstraction without replicating stored data.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain 1NF and 2NF with examples.

1NF: Eliminates repeating groups and non-atomic values.
2NF: In 1NF and contains no partial functional dependencies (all non-key attributes depend fully on the primary key).

Q7 (7.5M) **What is Query Optimization? Explain heuristic optimization.**

Explanation: Query optimization is the process of choosing the most efficient execution plan for evaluating a query. Heuristic optimization applies algebraic rewrite rules:

- Perform `Selection (σ)` as early as possible (reduces intermediate relation sizes).
- Perform `Projection (π)` early to eliminate unneeded attributes.
- Execute the most restrictive selections before performing Joins.

Q8 (7.5M) **Explain Log-Based Recovery with WAL (Write-Ahead Logging) and Checkpoints.**

Log-Based Recovery: Records all database modifications in an append-only log file before writing changes to disk (Write-Ahead Logging). During crash recovery, log records are used to UNDO incomplete transactions and REDO committed ones. Checkpoints reduce recovery overhead by flushing dirty buffers periodically.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain the Three-Level ANSI/SPARC Database Architecture with detailed diagram.**

Architecture: External Schema (Views) → Conceptual Schema (Logical tables) → Internal Schema (Physical disk blocks and access paths). Ensures physical and logical data independence.

Q10 (15M) **Given tables VENDOR(VCode, Name, Address, Phone) and PRODUCT(PCode, PName, Price, VCode). Write SQL queries for:**
(i) Display all vendors from 'Delhi'.
(ii) Display products with price greater than 500.
(iii) Display vendor name who supplies product 'Laptop'.
(iv) Count total products supplied by each vendor.

SQL Solutions:

```
-- (i) Vendors from Delhi
SELECT * FROM VENDOR WHERE Address LIKE '%Delhi%';

-- (ii) Products with price > 500
SELECT PCode, PName, Price FROM PRODUCT WHERE Price > 500;

-- (iii) Vendor supplying 'Laptop'
SELECT V.Name
FROM VENDOR V
JOIN PRODUCT P ON V.VCode = P.VCode
WHERE P.PName = 'Laptop';

-- (iv) Count total products per vendor
SELECT V.VCode, V.Name, COUNT(P.PCode) AS TotalProducts
FROM VENDOR V
LEFT JOIN PRODUCT P ON V.VCode = P.VCode
GROUP BY V.VCode, V.Name;
```

Q11 (15M) **Explain Relational Algebra vs Relational Calculus.**

Comparison: Relational Algebra is procedural (specifies operations step-by-step), while Relational Calculus (Tuple & Domain) is non-procedural/declarative (expresses what is required via mathematical predicate logic). Codd's Theorem proves they are equivalent in expressive power.

Q12 (15M) **Explain BCNF (Boyce-Codd Normal Form) and compare with 3NF.**

BCNF: Stricter form of 3NF where every determinant in a non-trivial functional dependency must be a super key. Eliminates anomalies that 3NF allows when there are overlapping composite candidate keys.

Q13 (15M) **Given tables: PLAY(PlayCode, PlayName, DirNum), DIRECTOR(DirNum, DirName, DirDOB). Write SQL queries to find director names and plays directed.**

SQL Solutions:

```
-- Director of play 'Hamlet'
SELECT D.DirName
FROM DIRECTOR D
JOIN PLAY P ON D.DirNum = P.DirNum
WHERE P.PlayName = 'Hamlet';

-- All plays directed by each director
SELECT D.DirName, P.PlayName
FROM DIRECTOR D
JOIN PLAY P ON D.DirNum = P.DirNum
ORDER BY D.DirName;
```

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SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Define Super Keys, Candidate Keys, and Primary Keys.**

Definitions:

- **Super Key:** A set of one or more attributes that uniquely identifies each tuple in a relation (may contain redundant attributes).
- **Candidate Key:** A minimal super key having no extraneous/redundant attributes.
- **Primary Key:** The specific candidate key chosen by the database designer as the principal tuple identifier. Cannot contain `NULL`.

Q2 (3M) **Explain the term "Data Independence".**

Definition: Data Independence is the capacity to change the schema at one level of the database system without having to alter the schema at the next higher level. It includes Physical Data Independence (modifying internal storage without affecting conceptual schema) and Logical Data Independence (modifying conceptual schema without altering external views).

Q3 (3M) **What is Hashing? Explain.**

Definition: Hashing is a direct addressing technique where a mathematical algorithm (hash function) calculates the exact physical block address of a record from its search key value, enabling O(1) constant-time retrieval.

Q4 (3M) **What do you mean by Relational Database? Describe.**

Definition: A **Relational Database** is a collection of data items organized into formal tables (relations) of rows and columns based on first-order predicate logic, where relations are linked together via primary and foreign key constraints.

Q5 (3M) **Briefly explain the terms Security and Authorization.**

Definitions:

- **Database Security:** Protection of the database against unauthorized access, malicious attacks, corruption, or destruction.
- **Authorization:** The mechanism that defines and enforces what actions (e.g., SELECT, INSERT, UPDATE, DROP) a validated user or role is permitted to perform on specific database objects.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **What is a Referential Integrity Constraint? Why do we need it? Explain with a suitable example.**

Explanation: Referential integrity requires that a foreign key value in a referencing table must match an existing primary key value in the referenced table, or be `NULL`.
Need: Prevents orphan records and inconsistent states.
Example:

```
-- Parent Table
CREATE TABLE Department (DeptNo INT PRIMARY KEY, DName VARCHAR(30));

-- Child Table with Foreign Key
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,
    EmpName VARCHAR(50),
    DeptNo INT,
    CONSTRAINT fk_emp_dept FOREIGN KEY (DeptNo) REFERENCES Department(DeptNo)
        ON DELETE CASCADE ON UPDATE CASCADE
);
```

If department 10 is deleted, `ON DELETE CASCADE` automatically removes all employees assigned to department 10, preventing dangling references.

Q7 (7.5M) **List four significant differences between a File-Processing System and a DBMS.**

Differences:

Aspect	File-Processing System	DBMS
Data Redundancy	High redundancy across duplicate files.	Controlled and normalized.
Data Independence	None (tight coupling with code).	Logical & Physical Data Independence.
Concurrency	Prone to race conditions and lost updates.	ACID compliant concurrency control.
Crash Recovery	Manual restore; inconsistent state.	Automated log-based rollback and redo.

Q8 (7.5M) **Explain Third Normal Form (3NF) with example.**

Definition: A relation is in **3NF** if it is in 2NF and has no transitive functional dependencies for non-prime attributes. Formally, for every non-trivial FD X \to Y, either X is a super key or Y is a prime attribute.
Example: `Student(RollNo, Name, DeptID, DeptHead)`.
RollNo \to DeptID and DeptID \to DeptHead (Transitive!).
Decompose into: `Student(RollNo, Name, DeptID)` and `Department(DeptID, DeptHead)`.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain the different locking techniques with examples for concurrency control.**

Locking Techniques:

1. **Shared Lock (S-Lock / Read Lock):** Multiple transactions can simultaneously hold shared locks to read data without conflict.
2. **Exclusive Lock (X-Lock / Write Lock):** Only one transaction can hold an exclusive lock to read and write. No other transaction can hold any lock on that item.
3. **Two-Phase Locking (2PL):** Growing Phase (acquiring locks) followed by Shrinking Phase (releasing locks). Guarantees conflict serializability.
4. **Strict 2PL:** Exclusive locks held until transaction commits, eliminating cascading rollbacks.

Q10 (15M) **Explain Indexed Sequential, B-Tree, and B+ Tree file organizations with examples.**

Comparison of File Organizations:

- **Indexed Sequential (ISAM):** Records stored in sequential key order with a multilevel static index. Good for sequential and direct access, but degrades with many overflow records.
- **B-Tree:** Self-balancing dynamic multi-way search tree. Keys and data record pointers reside in both internal nodes and leaf nodes.
- **B+ Tree:** Modern standard for relational DBMS storage engines. Internal nodes store only search keys to guide navigation, while **all actual data records/pointers reside in leaf nodes** connected via a doubly linked list, enabling superior range scans.

Q11 (15M) **Explain Super Class, Sub Class, Inheritance, and Generalization with examples.**

Enhanced ER (EER) Concepts:

- **Superclass:** A generic entity set that includes distinct subgroups (e.g., `PERSON`).
- **Subclass:** A specialized sub-entity set that inherits all attributes and relationships from its superclass, while adding specialized attributes (e.g., `STUDENT` with GPA, `TEACHER` with Rank).
- **Attribute Inheritance:** Subclass entities automatically inherit the primary key and all general attributes of their superclass.
- **Generalization:** Bottom-up synthesis where lower-level entity sets are combined into a generalized superclass entity.

Q12 (15M) **Write short notes on: (i) Strong and Weak Entity, (ii) Different Levels of Data Abstraction, (iii) Database Recovery Techniques.**

(i) Strong vs Weak Entity: A Strong entity has a primary key of its own. A Weak entity does not have a primary key and depends on the existence of an identifying strong entity set (identified by discriminator + strong entity PK).
(ii) Data Abstraction Levels: Physical level (low-level block storage), Logical level (community tables and relations), and View level (user-specific abstractions).
(iii) Database Recovery Techniques: Log-based recovery (Write-Ahead Logging), Deferred/Immediate update schemes, and Checkpoints to restore consistency following system failures.

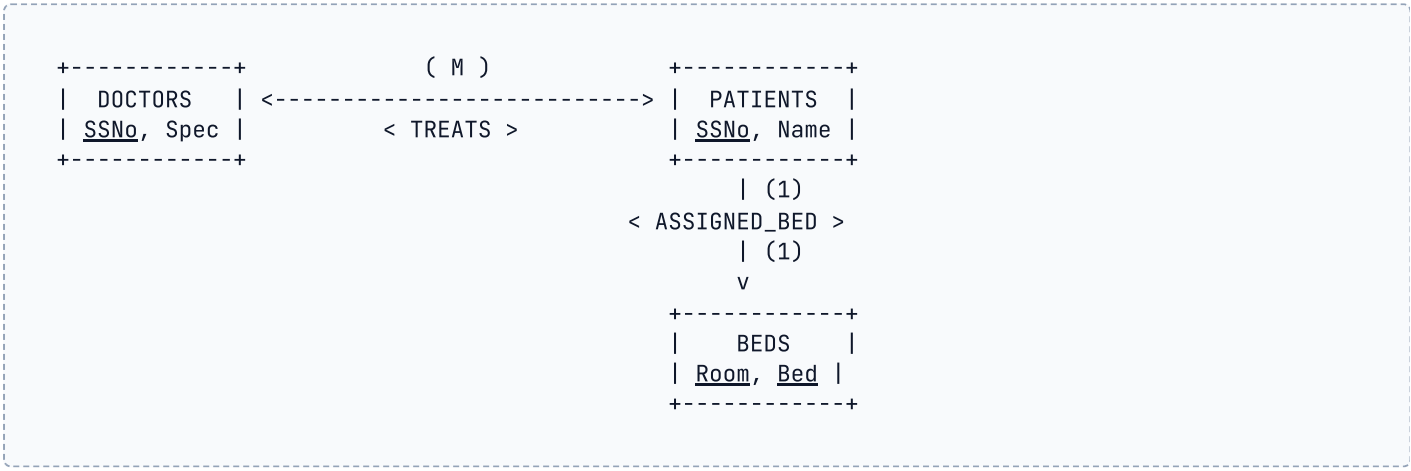
Q13 (15M) **A hospital maintains data about entities: PATIENTS, DOCTORS, BEDS, ACCOUNTS. Construct an E-R diagram specifying keys, mapping cardinalities, and participation constraints. Show table mapping.**

Entities & Attributes:

- `PATIENTS(SSNo, LastName, FirstName, HomePhone, Sex, DOB, Age, Street, City, State, Zip)`
- `DOCTORS(SSNo, LastName, FirstName, OfficePhone, Pager, Specialty)`
- `BEDS(RoomNumber, BedNumber, Type, Status, PricePerHour)`
- `ACCOUNTS(AccountNo, DateIn, DateOut, TotalAmount, Status)`

Relationships:

- `TREATS` (M:N between DOCTORS and PATIENTS)
- `ASSIGNED_BED` (1:1 between PATIENTS and BEDS)
- `HAS_ACCOUNT` (1:1 between PATIENTS and ACCOUNTS)



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SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) **Name the three levels of Data Abstraction.**

Three Levels:

1. **Physical Level (Lowest):** Describes how data is stored on disk blocks.
2. **Logical / Conceptual Level (Middle):** Describes what data is stored and relationships among data.
3. **View / External Level (Highest):** Describes customized views for specific user groups.

Q2 (3M) **Differentiate between Entity Integrity and Referential Integrity.**

Differences:

- **Entity Integrity:** States that no primary key attribute value can be `NULL`, ensuring every tuple can be uniquely identified.
- **Referential Integrity:** States that a foreign key value must either match an existing primary key value in the parent relation or be `NULL`.

Q3 (3M) **Differentiate between Candidate Key and Primary Key.**

Explanation: A relation may possess multiple candidate keys (minimal super keys). The database designer selects one candidate key as the **Primary Key**, while all remaining candidate keys are designated as **Alternate Keys**.

Q4 (3M) **List any four disadvantages of File Processing Systems.**

Disadvantages: Data redundancy and inconsistency, lack of data independence, concurrent access anomalies, and poor security enforcement.

Q5 (3M) **Distinguish between Functional Dependency and Multivalued Dependency.**

Comparison:

- **Functional Dependency (X → Y):** For any two tuples with identical X values, their Y values must be identical (single-valued dependency). Basis of 2NF, 3NF, BCNF.
- **Multivalued Dependency (X ↠ Y):** The presence of (x, y_1, z_1) and (x, y_2, z_2) implies the presence of (x, y_1, z_2) and (x, y_2, z_1), representing independent multi-valued facts. Basis of 4NF.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Explain Second Normal Form (2NF) and Third Normal Form (3NF) with examples.**

2NF: In 1NF and no partial dependencies (every non-prime attribute depends on the whole candidate key).
3NF: In 2NF and no transitive dependencies (for every non-trivial FD X \to Y, X is a super key or Y is prime).

Q7 (7.5M) **Compare B-Tree and B+ Tree Index Structures.**

Comparison:

Feature	B-Tree	B+ Tree
Data Storage	Keys and data records/pointers stored in both internal and leaf nodes.	Data pointers stored **only in leaf nodes**; internal nodes store only routing keys.
Leaf Node Linking	Leaf nodes are not linked.	Leaf nodes are connected via a **doubly linked list**.
Range Queries	Slow (requires in-order tree traversal).	Extremely fast (scan linked list across leaf nodes).
Node Capacity	Fewer keys per node due to data pointers.	High fan-out, fewer levels, reduced disk I/O.

Q8 (7.5M) **Explain Conflict Serializability and Testing for Serializability using Precedence Graph.**

Precedence Graph Method:

1. Create a node for every committed transaction T_i.
2. Draw a directed edge T_i \to T_j if T_i performs a conflicting operation before T_j on the same data item (Write-Read, Read-Write, or Write-Write).
3. If the graph contains **NO cycle**, the schedule is conflict serializable. Topological sort gives the equivalent serial schedule.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain the Three-Tier Client/Server Architecture of DBMS with Data Independence.**

Three-Tier Architecture:

- **Presentation Tier (User Interface):** Web browser or desktop client GUI (React, HTML).
- **Application Tier (Business Logic):** Web/Application server executing business validation and query processing (Java, Node.js, Python).
- **Database Tier (Data Storage):** Enterprise RDBMS managing database engine, query processing, transactions, and tables.

Provides enhanced security, scalability, and clean modular separation.

Q10 (15M) **Draw an ER Diagram for a Library Management System with full entities and relational schema.**

Entities: `BOOK`, `MEMBER`, `PUBLISHER`, `BORROWS`.

```
[MEMBER] ----- (M:N via BORROWS with IssueDate, ReturnDate) ----- [BOOK]
                                                                    | (N:1)
                                                                [PUBLISHER]
```

Q11 (15M) **Discuss SQL Characteristics and explain five Aggregate Functions with complete query examples.**

SQL Queries for Aggregate Functions:

```
-- 1. COUNT: Total employees in IT
SELECT COUNT(*) FROM Employee WHERE Dept = 'IT';

-- 2. SUM: Total payroll for Sales
SELECT SUM(Salary) FROM Employee WHERE Dept = 'Sales';

-- 3. AVG: Average salary in company
SELECT AVG(Salary) FROM Employee;

-- 4. MIN and MAX: Salary bounds
SELECT MIN(Salary) AS MinSalary, MAX(Salary) AS MaxSalary FROM Employee;
```

Q12 (15M) **Discuss Selection, Projection, and Join operators of Relational Algebra with detailed expressions.**

Detailed Operators:

- \sigma_{City='Noida'}(Customer) selects rows where City is Noida.
- \pi_{Name, Email}(Customer) projects distinct Name and Email attributes.
- Customer \bowtie_{Customer.ID = Order.CustID} Order performs equijoin between Customer and Order relations.

Q13 (15M) **Explain Database Recovery Techniques: Log-Based Recovery, Immediate Update, Deferred Update, and Checkpoints.**

Log-Based Recovery:

- **Deferred Update (NO-UNDO / REDO):** Updates are recorded only in the log until commit. During recovery, only committed transactions are REDONE; uncommitted transactions require NO UNDO.
- **Immediate Update (UNDO / REDO):** Updates can be written to disk before commit. Active uncommitted transactions at crash must be UNDONE; committed ones are REDONE.
- **Checkpoints:** Periodically flushes dirty log buffers to disk. Recovery scans backward only to the most recent checkpoint, dramatically shortening crash recovery time.

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BCA-501: Introduction to DBMS

Session: December 2020 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Define DML (Data Manipulation Language).

Definition: Data Manipulation Language (DML) provides constructs for accessing, inserting, updating, and querying stored database records. Classified into *Procedural DML* (user specifies what data and how to retrieve it) and *Non-Procedural / Declarative DML* (SQL, where user specifies only what data is needed).

Q2 (3M) Give examples of Simple and Composite Attributes of an entity.

Examples:

- **Simple (Atomic) Attribute:** Cannot be divided into subparts (e.g., `RollNo = 101`, `Age = 21`, `Salary = 50000`).
- **Composite Attribute:** Can be divided into smaller sub-components with independent meanings (e.g., `Address` divided into `HouseNo`, `Street`, `City`, `Pincode`; or `Name` into `FirstName`, `LastName`).

Q3 (3M) What do you mean by Referential Integrity Constraint?

Definition: Referential Integrity requires that any value of a foreign key attribute in a referencing relation must either match an existing primary key value in the referenced relation or be `NULL`. Enforces valid relationships and prevents "orphan records".

Q4 (3M) What is a Data Dictionary (System Catalog)?

Definition: A **Data Dictionary** is a reserved set of read-only system tables maintained by the DBMS that stores metadata (data about data), including schemas, table descriptions, column domains, integrity constraints, indexes, user privileges, and storage allocations.

Q5 (3M) Define Relational Schema.

Definition: A **Relational Schema** denotes the design structure of a relation, symbolized as R(A_1, A_2, \dots, A_n), where R is the relation name and A_i are attribute names defined over specific domains dom(A_i).

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) Explain Lossless-Join Decomposition and Dependency Preservation in relational design.

1. Lossless-Join Decomposition:
A decomposition of relation R into R_1 and R_2 is lossless if and only if natural join of the projections equals the original relation without generating spurious tuples:

```
R_1 ⋈ R_2 = R ⇔ (R_1 ∩ R_2 → R_1) OR (R_1 ∩ R_2 → R_2)
```

2. Dependency Preservation:
A decomposition preserves dependencies if the union of functional dependencies enforceable within individual decomposed relations covers all functional dependencies of the original relation: (F_1 \cup F_2)^+ = F^+.

Q7 (7.5M) **Compare 3NF and BCNF with an illustrative example.**

Key Differences:

- **3NF:** For every non-trivial FD X \to Y, either X is a super key *OR* Y is a prime attribute (part of any candidate key). Allows dependency preservation always.
- **BCNF:** For every non-trivial FD X \to Y, X **must strictly be a super key**. Eliminates all FD redundancy, but may not always preserve all functional dependencies.

Q8 (7.5M) **Draw an ER Diagram for a University Examination System with entities, attributes, and relationships.**

Entities: `STUDENT` (RollNo, Name), `COURSE` (CourseCode, Title), `EXAM_CENTER` (CenterID, Location), `ADMIT_CARD` (AdmitCardNo).

```
[STUDENT] ----- (1:1) ----- [ADMIT_CARD]
    | (M)
    |
 < ENROLLS >
    |
    v (N)
 [COURSE] ----- (N:1) ----- [EXAM_CENTER]
```

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Discuss Relational Algebra fundamental operations with examples.**

Operations:

1. **Selection (σ):** \sigma_{Marks \ge 60}(Student)
2. **Projection (π):** \pi_{RollNo, Name}(Student)
3. **Cartesian Product (×):** Student \times Course
4. **Union (∪):** SectionA \cup SectionB
5. **Set Difference (−):** AllStudents - PassedStudents
6. **Rename (ρ):** \rho_{S}(Student)

Q10 (15M) **Explain File Organization Techniques: Sequential, Indexed Sequential (ISAM), and Hashed File Organization.**

1. Sequential File Organization: Records stored consecutively ordered by a search key. Fast for full table scans, slow O(N) for insertions/random lookups.
2. Indexed Sequential Access Method (ISAM): Combines sequential ordering with hierarchical index blocks (Cylinder, Track index). Enables rapid binary search.
3. Hashed File Organization: Uses hash function on search key for direct bucket addressing. Offers O(1) lookups, but poor for range queries.

Q11 (15M) **Explain Database Security and Authorization. Discuss GRANT and REVOKE commands in SQL.**

SQL Security Commands:

```
-- Granting SELECT and INSERT privileges with delegation rights
GRANT SELECT, INSERT ON Employee TO user_ravi WITH GRANT OPTION;

-- Granting role-based access
CREATE ROLE clerk;
GRANT SELECT, UPDATE(Salary) ON Employee TO clerk;
GRANT clerk TO user_amit;

-- Revoking privileges
REVOKE INSERT ON Employee FROM user_ravi CASCADE;
```

Q12 (15M) **Explain Two-Phase Locking (2PL) and describe Deadlock Prevention and Detection mechanisms.**

Deadlock Handling:

- **Deadlock Prevention:** Uses transaction timestamps to avoid circular wait:
  - *Wait-Die (Non-preemptive):* Older waits for younger; younger dies immediately.
  - *Wound-Wait (Preemptive):* Older wounds (preempts) younger; younger waits for older.
- **Deadlock Detection:** Builds periodic *Wait-For Graph (WFG)*; cycles in the graph indicate deadlocks, resolved by aborting the lowest-cost victim transaction.

Q13 (15M) **Explain Views in SQL. What are the advantages of Views? When is a View updatable?**

Views in SQL: A **View** is a virtual table defined by an underlying stored SQL query.
Advantages: Data security (restricts column access), simplified queries, and logical data independence.
Updatable View Criteria:

- Query contains only one base table in `FROM` clause.
- No aggregate functions (`SUM` , `AVG` , `COUNT`).
- No `GROUP BY` , `HAVING` , `DISTINCT` , or `UNION` .
- Primary key of base table must be included in the view columns.

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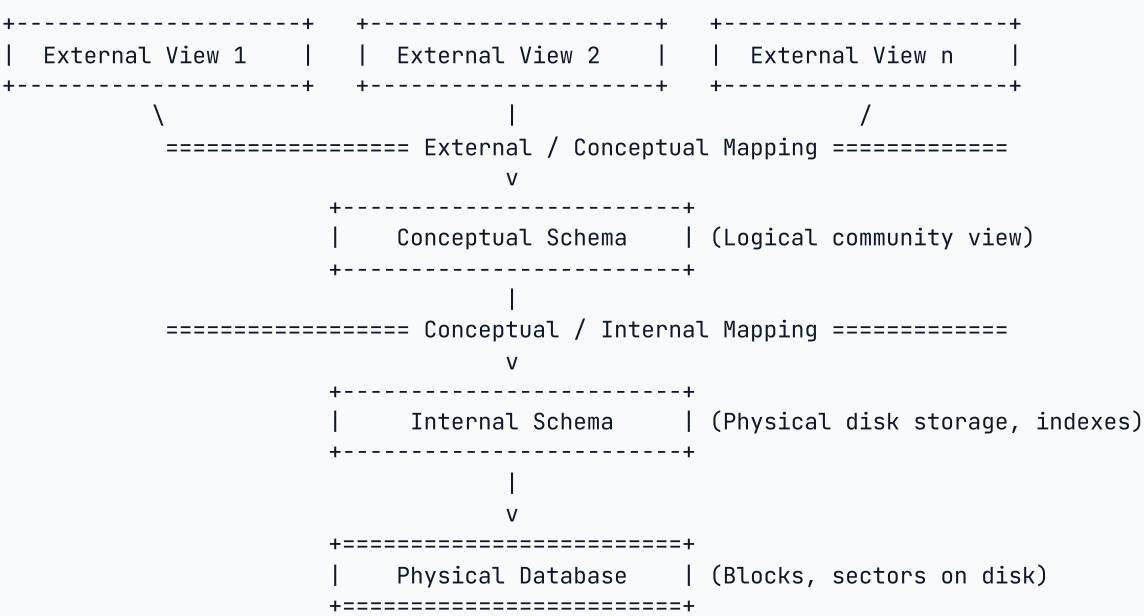
BCA-501: Introduction to DBMS

Session: December 2021 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks / CV-3 Pattern)

Q1 (7.5M) **Explain the Three-Level Architecture of DBMS with a neat block diagram.**

Three-Level ANSI/SPARC Architecture:



1. External Level (View Level): User-specific customized views presenting only relevant subsets of data while hiding confidential fields.
2. Conceptual Level (Logical Level): Describes *what data* is stored in the entire database and the relationships among data without physical storage details.
3. Internal Level (Physical Level): Describes *how data* is physically stored on storage media (record formats, B+ trees, hashing).

Q2 (7.5M) **Define Primary Key, NOT NULL Key, and UNIQUE Key with differences.**

Definitions & Differences:

Feature	PRIMARY KEY	UNIQUE KEY	NOT NULL
Purpose	Uniquely identifies each record.	Ensures distinct non-duplicate values.	Prevents missing or null values.
NULL Allowed	**Never** allowed (strictly 0 nulls).	Allowed (usually 1 or more nulls per column).	Does not enforce uniqueness, only disallows null.
Count per table	Only **1** per table.	Multiple unique constraints allowed.	Can be applied to every column.
Index created	Automatically creates Primary Clustered Index.	Creates Secondary Unique Non-Clustered Index.	No index created by default.

Q3 (7.5M) **Explain Hashing in brief with its collision handling techniques.**

Definition: Hashing is an efficient direct search technique where a mathematical hash function h(k) maps a search key k directly to the disk address/bucket number where the record resides, providing average O(1) time complexity.
Collision: Occurs when two distinct search keys hash to the same bucket address: h(k_1) = h(k_2).
Collision Handling Techniques:

- **Open Addressing (Linear Probing):** Searches the next contiguous sequential bucket until an empty slot is located.
- **Chaining / Separate Overflow Buckets:** Maintains a linked list of overflow buckets chained to the primary bucket.
- **Extendible Hashing (Dynamic Hashing):** Uses a directory of pointers with local and global depths to double directories and split buckets dynamically without rehashing all records.

Q4 (7.5M) **What are the pitfalls of Lock-Based Protocols in concurrency control?**

Pitfalls:

1. **Deadlock:** Occurs when two or more transactions are waiting indefinitely for locks held by each other (e.g., T_1 holds Lock on A and waits for B, while T_2 holds Lock on B and waits for A). Requires deadlock detection (Wait-For Graph) and victim rollback.
2. **Starvation:** A transaction waits indefinitely because other higher-priority transactions repeatedly acquire locks on the required data item.
3. **Cascading Rollback:** In non-strict 2PL, if a transaction aborts, all dependent uncommitted transactions that read its dirty data must also be rolled back in a cascade.
4. **Concurrency Bottleneck:** Excessive long-duration exclusive locks reduce system throughput and degrade response times.

Q5 (7.5M) **Why is Concurrency Control needed in DBMS? Explain with concurrency anomalies.**

Need for Concurrency Control: When multiple transactions execute concurrently without coordination, interference causes severe anomalies:

- **Lost Update Problem:** Two transactions overwrite each other's updates without reading the intermediate result.
- **Dirty Read (Uncommitted Dependency):** A transaction reads modified data from an uncommitted transaction that subsequently aborts.
- **Inconsistent Analysis (Non-Repeatable Read):** A transaction reads data while another transaction modifies it midway, producing skewed sums.

SECTION B — SHORT ANSWER QUESTIONS (1 × 15 = 15 Marks / CV-3 Pattern)

Q6 (15M) **Given the Relational Schema:**
EMPLOYEE(EmpID, EmpName, Age, Gender, Salary, City, DeptID, Designation, HireDate)
DEPARTMENT(DeptID, DeptName, Location)
Write SQL Queries for:
1. Display the age of 'Male' employees.
2. Display the name of the highest salary employee.
3. Display the name of department of employee 'Aman'.
4. Display all employees belonging to the same address (city).
5. Which employee is the oldest manager in the company?

SQL Queries:

```
-- Query 1: Age of Male employees
SELECT EmpName, Age
FROM EMPLOYEE
WHERE Gender = 'Male';

-- Query 2: Name of the highest salary employee
SELECT EmpName, Salary
FROM EMPLOYEE
WHERE Salary = (SELECT MAX(Salary) FROM EMPLOYEE);

-- Query 3: Department name of employee 'Aman'
SELECT D.DeptName
FROM DEPARTMENT D
JOIN EMPLOYEE E ON D.DeptID = E.DeptID
WHERE E.EmpName = 'Aman';

-- Query 4: Display all employees grouped by identical city
SELECT City, COUNT(EmpID) AS TotalEmps,
       GROUP_CONCAT(EmpName) AS EmployeeNames
FROM EMPLOYEE
GROUP BY City
HAVING COUNT(EmpID) > 1;

-- Query 5: Oldest manager in the company
SELECT EmpName, Age, Designation, HireDate
FROM EMPLOYEE
WHERE Designation = 'Manager'
ORDER BY Age DESC
LIMIT 1;
```

SECTION C — DETAILED ANSWER QUESTIONS (2 × 22.5 = 45 Marks / CV-3 Pattern)

Q7 (22.5M) **Explain Two-Phase Locking (2PL) Protocol. Compare Strict 2PL and Rigorous 2PL with serializability guarantees.**

Two-Phase Locking (2PL): A concurrency control protocol guaranteeing conflict serializability by dividing a transaction into two distinct phases:

1. **Growing Phase:** Transaction may obtain locks (Shared or Exclusive) but cannot release any lock.
2. **Shrinking Phase:** Transaction may release locks but cannot acquire any new lock.

Variants of 2PL:

- **Basic 2PL:** Guaranteed conflict serializability, but vulnerable to cascading aborts and deadlocks.
- **Strict 2PL:** All Exclusive (X) locks held by a transaction must be retained until the transaction commits or aborts. *Guarantees strict schedules and eliminates cascading aborts!*
- **Rigorous 2PL:** Both Shared (S) and Exclusive (X) locks must be retained until transaction commit/abort. Ensures serializability in commit order.

Q8 (22.5M) **Define Normalization. Explain 1NF, 2NF, 3NF, and BCNF with a complete enterprise example.**

Comprehensive Enterprise Example:
Consider unnormalized table: `STUDENT_COURSE_REG(StudentID, StudentName, CourseID, CourseName, Instructor, InstructorOffice)`.
Functional Dependencies:

- StudentID \to StudentName
- CourseID \to CourseName, Instructor
- Instructor \to InstructorOffice

Candidate Key: (StudentID, CourseID).

Step 1: Convert to 2NF (Eliminate Partial Dependencies):
- `STUDENT(StudentID, StudentName)`
- `COURSE(CourseID, CourseName, Instructor, InstructorOffice)`
- `ENROLLMENT(StudentID, CourseID)`

Step 2: Convert to 3NF (Eliminate Transitive Dependency in COURSE):
In `COURSE`, Instructor \to InstructorOffice is transitive.
- `COURSE(CourseID, CourseName, Instructor)` (FK: Instructor)
- `INSTRUCTOR(Instructor, InstructorOffice)`

All relations are now in 3NF and BCNF with zero redundancy!

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BCA-501: Introduction to DBMS

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SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) Define Database Management System (DBMS).

Definition: A **Database Management System (DBMS)** is a computerized system software suite that enables users to define, create, maintain, query, and control access to a shared database. It serves as an intelligent interface between end-user applications and physical data storage.

Q2 (3M) List any five applications of DBMS.

Applications:

1. **Banking:** Customer accounts, loans, ATM transactions, fund transfers.
2. **Airlines & Railways:** Real-time reservation systems, flight schedules, ticketing.
3. **Universities & Colleges:** Student admissions, course registrations, grades, fee accounts.
4. **E-Commerce & Retail:** Online product catalogs, shopping carts, billing, inventory control.
5. **Telecommunications:** Call records, monthly billing, subscriber information, network routing.

Q3 (3M) What are the disadvantages of a File Processing System?

Disadvantages:

- **Data Redundancy & Inconsistency:** Same data stored in multiple files leading to conflicting copies.
- **Difficulty in Accessing Data:** Requires writing a new custom program for every ad-hoc query.
- **Data Isolation:** Data scattered across disparate file formats cannot be easily joined.
- **Integrity Problems:** Hard to enforce consistency constraints across independent files.
- **Atomicity Failures:** Incomplete updates during crashes leave data in corrupt states.

Q4 (3M) Define Data Model.

Definition: A **Data Model** is an abstract framework providing a collection of conceptual concepts for describing data, relationships, semantics, and constraints in a database. Major categories: Conceptual/High-level (ER Model), Representational/Logical (Relational Model), and Physical/Low-level models.

Q5 (3M) What are Attributes? Give examples.

Definition: An **Attribute** is a descriptive property or characteristic that describes an entity in an entity set.
Examples for Entity STUDENT:

- *Simple Attribute:* `RollNo`, `Age`
- *Composite Attribute:* `FullName` (composed of `FirstName`, `LastName`)
- *Multivalued Attribute:* `PhoneNumbers`, `Hobbies`
- *Derived Attribute:* `Age` (computed from `DateOfBirth` and current date)

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Compare Generalization and Specialization with examples.**

Comparison:

Criteria	Generalization	Specialization
Approach	**Bottom-up** design process	**Top-down** design process
Mechanism	Combines common features of multiple lower-level entities into a single higher-level entity.	Splits a higher-level entity into specialized sub-entities based on distinct features.
Goal	Minimizes redundancy by factoring out common attributes.	Identifies distinguishing characteristics of specific sub-categories.
Example	Entities `CAR` and `TRUCK` generalized into higher entity `VEHICLE`.	Entity `EMPLOYEE` specialized into `SALARIED_EMP` and `HOURLY_EMP`.

Q7 (7.5M) **Explain Primary Key, Super Key, and Foreign Key with examples.**

Definitions & Relationships:

- **Super Key:** A set of one or more attributes that uniquely identifies a tuple within a relation. May contain redundant attributes (e.g., `{EmpID, Name}`).
- **Primary Key:** A strictly minimal candidate key selected by the database designer to uniquely identify each tuple. No duplicates, no `NULL` allowed.
- **Foreign Key:** An attribute that establishes a referential relationship by referencing the primary key of another table.

```
Relation: EMPLOYEE(EmpID, EmpName, Email, DeptNo)
Relation: DEPARTMENT(DeptNo, DeptName, Building)

Super Keys: {EmpID}, {EmpID, EmpName}, {Email}, {EmpID, Email}
Candidate Keys: {EmpID}, {Email}
Primary Key: {EmpID}
Foreign Key: {DeptNo} in EMPLOYEE references DeptNo in DEPARTMENT
```

Q8 (7.5M) **What is Normalization? Explain the importance of Normalization in database design.**

Definition: Normalization is the systematic process of organizing relation schemas based on functional dependencies to eliminate anomalies and avoid redundant storage.

Importance:

- Eliminates Insertion, Deletion, and Modification anomalies.
- Reduces storage space by ensuring each fact is recorded in only one place.
- Prevents data corruption by enforcing referential integrity constraints across smaller, well-structured tables.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

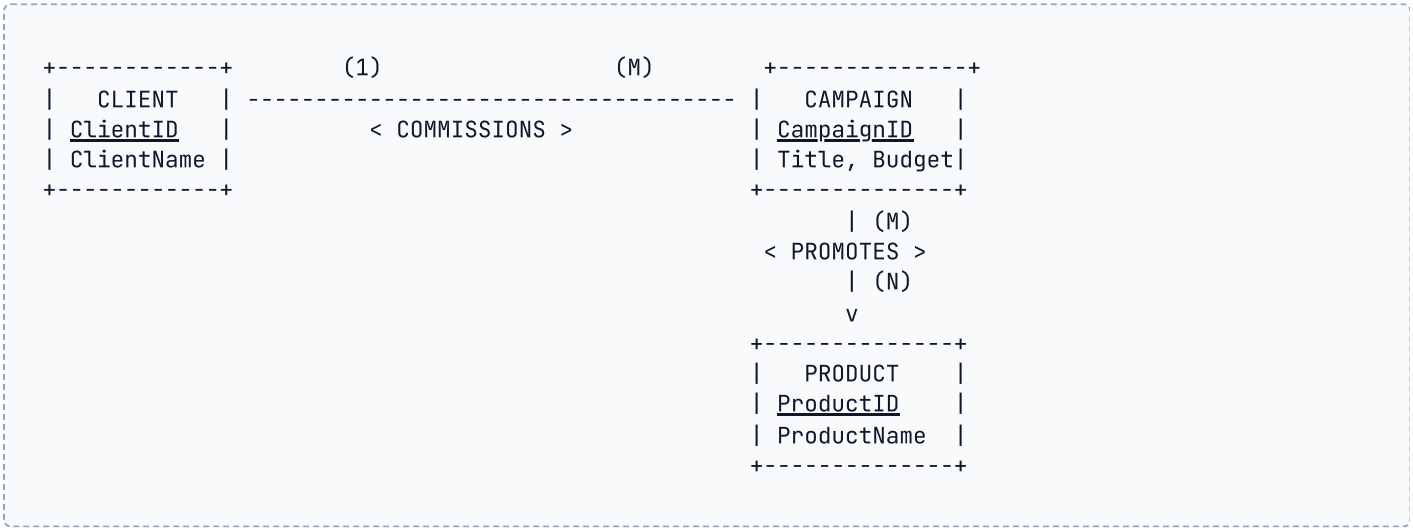
Q9 (15M) **Draw an ER Diagram for a small Marketing Company database, assuming your own data requirements.**

Enterprise Requirements:

- **CAMPAIGN:** CampaignID, Title, Budget, StartDate, EndDate.
- **CLIENT:** ClientID, ClientName, ContactPerson, Email, Phone.
- **PRODUCT:** ProductID, ProductName, Category.
- **MARKETING_AGENT:** AgentID, AgentName, CommissionRate.

Relationships:

- A Client commissions one or more Campaigns (1:N).
- Each Campaign promotes one or more Products (M:N).
- Agents are assigned to Campaigns (M:N).



Q10 (15M) **Describe the classification of Database Languages. Which type of language is SQL?**

Classification of Database Languages:

1. **Data Definition Language (DDL):** Specifies the database schema, storage structure, and constraints (`CREATE` , `ALTER` , `DROP` , `TRUNCATE`).
2. **Data Manipulation Language (DML):** Retrieves, inserts, updates, and deletes stored data. Can be procedural (relational algebra) or non-procedural/declarative (relational calculus, SQL).
3. **Data Control Language (DCL):** Controls permissions and access authorization (`GRANT` , `REVOKE`).
4. **Transaction Control Language (TCL):** Manages the logical unit of work (`COMMIT` , `ROLLBACK` , `SAVEPOINT`).

Classification of SQL: SQL is a **comprehensive declarative (non-procedural) language** that integrates DDL, DML, DCL, and TCL in a single standardized unified language.

Q11 (15M) **Explain DDL Commands and DML Commands with complete SQL syntax and executable examples.**

1. DDL Commands (Data Definition Language):

```
-- 1. CREATE TABLE
CREATE TABLE Student (
    RollNo INT PRIMARY KEY,
    Name VARCHAR(50) NOT NULL,
    Branch VARCHAR(20) DEFAULT 'BCA',
    FeePaid DECIMAL(10,2) CHECK (FeePaid ⩾ 0)
);

-- 2. ALTER TABLE: Add and modify column
ALTER TABLE Student ADD COLUMN Email VARCHAR(100);
ALTER TABLE Student MODIFY COLUMN Name VARCHAR(80);

-- 3. DROP & TRUNCATE
TRUNCATE TABLE Student; -- Deletes all rows, preserves table structure
DROP TABLE Student;     -- Deletes rows and schema permanently
```

2. DML Commands (Data Manipulation Language):

```
-- 1. INSERT
INSERT INTO Student (RollNo, Name, Branch, FeePaid, Email)
VALUES (101, 'Aman Sharma', 'BCA', 25000.00, 'aman@example.com');

-- 2. UPDATE
UPDATE Student
SET FeePaid = 30000.00
WHERE RollNo = 101;

-- 3. DELETE
DELETE FROM Student WHERE RollNo = 101;

-- 4. SELECT
SELECT RollNo, Name, FeePaid FROM Student WHERE Branch = 'BCA';
```

Q12 (15M) **Explain Data Independence with its types: Physical and Logical Data Independence. Why is it significant?**

Definition: Data Independence is the ability to modify schema definitions at one level of the 3-level ANSI/SPARC architecture without altering schemas at the higher level.

1. Logical Data Independence:

- The ability to modify the **Conceptual Schema** (e.g., adding/deleting attributes, splitting tables) without having to rewrite external schemas or existing application programs.
- *How achieved:* Using SQL Views that present the original logical schema to old applications.
- *Difficulty:* Harder to achieve because applications are closely tied to logical data models.

2. Physical Data Independence:

- The ability to modify the **Internal/Physical Schema** (e.g., changing file organizations, building B+ Tree indexes, moving to new storage media) without changing the Conceptual Schema.
- *Ease:* Easier to achieve because logical structure remains completely decoupled from disk-level storage.

Q13 (15M) Who are Data Administrators (DA)? What are the functions of a Database Administrator (DBA)? Differentiate between DA and DBA.

Data Administrator (DA) vs Database Administrator (DBA):

Aspect	Data Administrator (DA)	Database Administrator (DBA)
Role Level	Managerial and business level	Technical and operational level
Focus	Business data policies, enterprise data dictionary, data governance.	Physical database design, database server installation, query tuning, security.
Scope	Decides *what data* needs to be collected across the enterprise.	Decides *how data is stored, indexed, secured, and recovered*.

Key Functions of DBA:

1. Hardware/software installation and database upgrades.
2. Creating database users, granting roles, and monitoring access privileges.
3. Performance tuning: Analyzing execution plans, optimizing SQL queries, re-indexing.
4. Automating database backups and executing disaster recovery drills.

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BCA-501: Introduction to DBMS

Session: December 2024 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is Schema in DBMS?

Definition: A **Database Schema** is the formal structural blueprint and design description of the entire database. It specifies:

- Table names, columns, data types, and domains.
- Integrity constraints (primary keys, foreign keys, check constraints).
- Relationships between entities and storage representations.

It is stored in the system catalog and does not change frequently.

Q2 (3M) What are the advantages of using DBMS?

Key Advantages:

1. **Minimal Redundancy:** Normalization ensures data is stored in one primary location.
2. **Data Integrity:** Automatic enforcement of domain and referential integrity constraints.
3. **Concurrency Control:** Multi-user simultaneous access without collisions via locking protocols.
4. **Crash Recovery:** WAL (Write-Ahead Logging) and checkpoints guarantee durability after hardware failure.
5. **Data Independence:** Separates user views from low-level storage modifications.

Q3 (3M) Who is a DBA? Mention their key functions.

Definition: A **Database Administrator (DBA)** is an IT professional who manages the database environment, maintains performance, controls user permissions, schedules backups, and enforces security policies.

Q4 (3M) What is a Database?

Definition: A **Database** is an integrated and logically related collection of records, designed to meet the information processing needs of multiple users across an organization.

Q5 (3M) What is RDBMS?

Definition: A **Relational Database Management System (RDBMS)** is a DBMS based on E.F. Codd's Relational Model (1970). Data is organized into 2D tables (relations) consisting of rows (tuples) and columns (attributes), manipulated using SQL and relational algebra. Examples: Oracle, MySQL, PostgreSQL, SQL Server.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Find the minimum number of tables required for the given ER diagram in the relational model. Explain table conversion rules.**

Rules for ER to Relational Reduction:

1. **Strong Entity Sets:** Each strong entity converts into a separate independent table.
2. **Weak Entity Sets:** Converts into a table containing its discriminator (partial key) combined with the primary key of the identifying strong entity as composite primary key.
3. **1:1 Relationships:** Combine both entities into 1 table, or place the primary key of one table as a foreign key in the other table.
4. **1:N Relationships:** Place the primary key of the '1' side entity into the table of the 'N' side entity as a foreign key (no new table needed).
5. **M:N Relationships:** Requires a new separate relationship table containing primary keys of both participating entities as foreign keys.

Result: For an ER model with two strong entities A and B in 1:N relationship, exactly **2 tables** are needed. If the relationship is M:N, exactly **3 tables** are required.

Q7 (7.5M) **What are UNION, MINUS, and INTERSECT commands in DBMS? Explain with syntax and examples.**

Conditions: Both queries must be *Union-Compatible* (same number of columns with compatible data types).

- `UNION` : Combines results of two SELECT queries and eliminates duplicate tuples.
- `INTERSECT` : Returns only the common tuples present in both result sets.
- `MINUS / EXCEPT` : Returns tuples from the first query that are not present in the second query.

SQL Examples:

```
-- UNION: All students from CS or IT
SELECT RollNo, Name FROM Student_CS
UNION
SELECT RollNo, Name FROM Student_IT;

-- INTERSECT: Students enrolled in both Math and Physics clubs
SELECT StudentID FROM MathClub
INTERSECT
SELECT StudentID FROM PhysicsClub;

-- MINUS: Registered students who have NOT paid exam fees
SELECT StudentID FROM RegisteredStudents
EXCEPT
SELECT StudentID FROM PaidStudents;
```

Q8 (7.5M) **Define the following terms: (a) Data Model, (b) Database Schema.**

(a) Data Model: A conceptual collection of tools, constructs, and notations used to describe data, data relationships, data semantics, and consistency constraints. Major models include:

- **Relational Model:** Tables of tuples and attributes (SQL).
- **Entity-Relationship (ER) Model:** High-level conceptual modeling.
- **Object-Oriented & Document (NoSQL) Models:** JSON, key-value, graph.

(b) Database Schema: The overall structural skeleton of the database defined during the design phase, distinct from the dynamic instances/states stored inside it.

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Define Normal Forms. Explain 1NF, 2NF, 3NF, and BCNF with suitable examples.**

Detailed Normal Forms Analysis:

Normal Form	Requirement / Condition	Anomaly Eliminated
1NF	All attribute values must be atomic (no multivalued / nested attributes).	Repeating groups and unstructured data arrays
2NF	In 1NF and no non-prime attribute is partially dependent on any candidate key.	Partial Functional Dependencies
3NF	In 2NF and for every X → Y, either X is a Super Key or Y is a Prime Attribute.	Transitive Functional Dependencies
BCNF	For every non-trivial functional dependency X → Y, X must strictly be a Super Key.	All functional dependency based redundancies

BCNF Example:
Suppose `STUDENT_ADVISOR(StudentID, Subject, Advisor)` where each student has an advisor per subject, and each advisor teaches only one subject: FD_1: (StudentID, Subject) \to Advisor, FD_2: Advisor \to Subject.
Candidate Key is (StudentID, Subject) or (StudentID, Advisor).
In FD_2, `Advisor` is not a super key, violating BCNF.
Decompose into: `Advisor_Subject(Advisor, Subject)` and `Student_Advisor(StudentID, Advisor)`. Both are now in BCNF!

Q10 (15M) **Explain Normalization with its advantages. What anomalies arise in unnormalized databases?**

Anomalies in Unnormalized Tables:

1. **Insertion Anomaly:** Cannot record information about an entity without having to insert another unrelated entity (e.g., cannot register a new department until at least one employee is hired).
2. **Deletion Anomaly:** Deleting a record of one entity unintentionally erases information about another entity (e.g., deleting the last employee in a department deletes all department details).
3. **Update / Modification Anomaly:** Changing an entity's attribute requires updating hundreds of duplicate rows; if one row is missed, data becomes contradictory.

Advantages of Normalization:

- Eliminates data redundancy and conserves disk storage.
- Maintains referential integrity automatically across tables.
- Speeds up row updates, inserts, and index lookups.

Q11 (15M) **Explain SQL with its various characteristics and features.**

Structured Query Language (SQL): The ANSI/ISO standard declarative language for relational databases.
Key Characteristics:

- **Declarative / Non-Procedural:** Users specify *what data* they want, leaving the query optimizer to determine *how* to retrieve it.
- **Comprehensive Capabilities:** Encompasses DDL (structure), DML (manipulation), DCL (security/privileges), and TCL (transactions).
- **Portability & Standards:** Works uniformly across major vendors (Oracle, MySQL, SQL Server, PostgreSQL).
- **Integrity Constraints Support:** Native enforcement of `PRIMARY KEY`, `FOREIGN KEY`, `CHECK`, `UNIQUE`, and `DEFAULT`.
- **View & Subquery Support:** Enables nested queries, common table expressions (CTEs), and dynamic virtual tables (Views).

Q12 (15M) **What is Relational Algebra? Discuss its basic and extended operations with expressions.**

Overview: Relational algebra provides the formal theoretical mathematical foundation for query evaluation and execution plan optimization in RDBMS.
Operations Summary:

- **Selection (σ):** Horizontal subset of rows based on conditions.
- **Projection (π):** Vertical subset of columns, duplicate-free.
- **Cartesian Product (×):** All pairs of concatenated rows.
- **Set Union (∪), Intersection (∩), Difference (−):** Set-theoretic operations on union-compatible relations.
- **Theta Join ($\bowtie_\theta$):** Cross product followed by condition filtering.
- **Division (÷):** Used for queries involving "for all" / universal quantification.

Q13 (15M) **Explain the reduction of ER Schema to Relational Tables with a complete case study.**

Case Study: Company Database:
Entities: `EMPLOYEE`, `DEPARTMENT`, `PROJECT`.
Relationships:

- `WORKS_FOR` (N:1 from Employee to Department)
- `MANAGES` (1:1 from Employee to Department with `StartDate`)
- `WORKS_ON` (M:N between Employee and Project with `Hours`)

Relational Reduction:

```
1. DEPARTMENT(DeptNo, DeptName, MgrEmpID, MgrStartDate)
   -- 1:1 MANAGES folded into DEPARTMENT with MgrEmpID as FK
2. EMPLOYEE(EmpID, EmpName, Salary, DeptNo)
   -- N:1 WORKS_FOR represented via DeptNo as FK referencing DEPARTMENT
3. PROJECT(ProjNo, ProjName, DeptNo)
   -- N:1 CONTROLS represented via DeptNo as FK
4. WORKS_ON(EmpID, ProjNo, Hours)
   -- M:N relationship becomes a separate junction table with composite PK
   -- FK EmpID → EMPLOYEE, FK ProjNo → PROJECT
```

CHAUDHARY CHARAN SINGH UNIVERSITY (CCSU) • BCA V SEMESTER

BCA-501: Introduction to DBMS

Session: December 2025 Examination | **Paper Code:** 18021 | **Authentic University Solved Paper (All Questions Solved)** | **Time:** 3 Hours • **Max Marks:** 75

SECTION A — VERY SHORT ANSWER QUESTIONS (5 × 3 = 15 Marks)

Q1 (3M) What is a Database?

Definition: A **Database** is an organized, logically coherent collection of related data modeled to represent real-world entities and enterprise operations. Key properties include:

- **Self-describing nature:** Stores both data and metadata (system catalog/data dictionary).
- **Program-data independence:** Structure of data files is stored separately from user access programs.
- **Multi-user transaction processing:** Guarantees concurrent access without collision or corruption.

Q2 (3M) State the key advantages of DBMS over File Processing Systems.

Advantages:

Aspect	Traditional File System	Database Management System (DBMS)
Data Redundancy	High redundancy across duplicate files	Controlled and minimized via normalization
Data Inconsistency	High risk due to isolated updates	Eliminated via centralized repository
Data Independence	None; changes in files break code	Physical and Logical Data Independence achieved
Concurrent Access	Prone to race conditions and lost updates	ACID compliant concurrency control (2PL, timestamp)
Security & Integrity	OS-level passwords, hard to enforce	Fine-grained role-based access & constraints

---

Q3 (3M) Who is a Database Administrator (DBA)? Mention key duties.

Definition: A **Database Administrator (DBA)** is the central authority responsible for supervising and administering the entire database environment.
Primary Functions:

1. **Schema Definition:** Designing and altering conceptual, external, and internal schemas.
2. **Storage & Access Tuning:** Specifying storage structures, file organizations, and primary/secondary indexes.
3. **Security & Authorization:** Granting and revoking user privileges, managing roles, and enforcing access control.
4. **Backup & Recovery:** Scheduling periodic backups and executing disaster recovery strategies.
5. **Integrity Enforcement:** Defining domain, entity, and referential integrity constraints.

Q4 (3M) Differentiate between Schema and Instance in DBMS.

Comparison:

- **Database Schema:** The overall logical blueprint, structure, and design of the database (e.g., table names, column data types, constraints). It is specified during system design and changes very infrequently.
- **Database Instance (State):** The actual data content populated within the database at a specific moment in time. It changes dynamically whenever records are inserted, updated, or deleted.

Analogy: In programming languages, a Schema corresponds to a Class/Data Type, whereas an Instance corresponds to an Object/Value of that type.

Q5 (3M) **What do you mean by Data Inconsistency in File Management Systems? Explain.**

Explanation: Data Inconsistency occurs when multiple copies of the exact same data across different files contain conflicting, contradictory values.

Example: In a university file system, a student's address is recorded in the *Library File* and also in the *Fee Accounting File*. If the student changes their residential address and only the Library office updates their local file while the Fee office does not, the university now holds two contradictory addresses for the same student. This occurs due to uncoordinated duplicate storage without centralized integrity control.

SECTION B — SHORT ANSWER QUESTIONS (2 × 7.5 = 15 Marks)

Q6 (7.5M) **Explain ER Diagram. Describe all standard ER notations and symbols with diagrams.**

Definition: An **Entity-Relationship (ER) Diagram** is a graphical semantic modeling tool that represents the logical structure of a database in terms of entities, their attributes, and relationships.

Standard Symbols & Notations:

```
[ Rectangle ]           → Entity Set (e.g., STUDENT, COURSE)
[[ Double Rectangle ]]  → Weak Entity Set (e.g., DEPENDENT, INSTALLMENT)
( Ellipse / Oval )      → Attribute (e.g., Age, Name)
( Underlined Text )     → Primary Key Attribute (e.g., RollNo)
(( Double Ellipse ))    → Multivalued Attribute (e.g., PhoneNumbers, Skills)
( Dashed Ellipse )      → Derived Attribute (e.g., Age calculated from DOB)
{ Diamond }             → Relationship Set (e.g., ENROLLED_IN)
{{ Double Diamond }}    → Identifying Relationship for Weak Entity Set
======== Double Line    → Total Participation (Every entity must participate)
-------- Single Line    → Partial Participation
```

Q7 (7.5M) **Explain Primary Key, Foreign Key, and Candidate Key with suitable SQL examples.**

1. Candidate Key: A minimal super key — an attribute or minimal set of attributes that uniquely identifies each tuple in a relation without any redundant attributes.

2. Primary Key: The specific candidate key chosen by the database designer to uniquely identify tuples in a table. It cannot accept duplicate values and cannot contain `NULL`.

3. Foreign Key: An attribute in a child table whose values must match the primary key of a referenced parent table (or be `NULL`), enforcing referential integrity.

SQL Implementation Example:

```sql
-- Parent Table: DEPARTMENT
CREATE TABLE Department (
    DeptID INT PRIMARY KEY,              -- Primary Key
    DeptName VARCHAR(50) NOT NULL UNIQUE -- Candidate Key
);

-- Child Table: EMPLOYEE
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,               -- Primary Key
    EmpName VARCHAR(100) NOT NULL,
    Email VARCHAR(100) UNIQUE,           -- Candidate Key
    DeptID INT,                          -- Foreign Key
    CONSTRAINT fk_dept FOREIGN KEY (DeptID) REFERENCES Department(DeptID)
        ON DELETE CASCADE ON UPDATE CASCADE
);
```

Q8 (7.5M) **Differentiate between DDL and DML commands with syntax and examples.**

Comparison:

Feature	Data Definition Language (DDL)	Data Manipulation Language (DML)
Purpose	Defines and modifies the schema structure of the database.	Manages, accesses, and manipulates the data within the tables.
Commit Nature	Auto-committed permanently (cannot be rolled back in most RDBMS).	Transactional (can be committed or rolled back using TCL).
Commands	`CREATE`, `ALTER`, `DROP`, `TRUNCATE`, `RENAME`	`INSERT`, `UPDATE`, `DELETE`, `SELECT`

SQL Examples:

```
-- DDL Commands:
CREATE TABLE Student (RollNo INT PRIMARY KEY, Name VARCHAR(50));
ALTER TABLE Student ADD COLUMN City VARCHAR(30);

-- DML Commands:
INSERT INTO Student (RollNo, Name, City) VALUES (101, 'Aman', 'Meerut');
UPDATE Student SET City = 'Delhi' WHERE RollNo = 101;
```

SECTION C — DETAILED ANSWER QUESTIONS (3 × 15 = 45 Marks)

Q9 (15M) **Explain the ACID properties of a transaction with practical examples. Why are they essential?**

Definition: A **Transaction** is a logical unit of database processing that includes one or more database access operations. To preserve database consistency and reliability, every DBMS must guarantee the four **ACID Properties**:

1. Atomicity ("All or Nothing"):

- Either all operations of the transaction are committed successfully, or none take effect. If failure occurs mid-transaction, all changes are rolled back.
- *Example:* Bank transfer of \$500 from Account A to Account B. If \$500 is debited from A and power fails before crediting B, the recovery subsystem rolls back A's debit.

2. Consistency (Preservation of Invariants):

- Execution of a transaction in isolation preserves database integrity constraints (e.g., total balance of A + B must remain unchanged before and after the transfer).

3. Isolation (Independence of Concurrent Executions):

- Even if multiple transactions execute concurrently, the intermediate uncommitted state of one transaction is hidden from others, producing a serializable execution schedule. Enforced via locking (2PL) and timestamp ordering.

4. Durability (Survivability of Committed Data):

- Once a transaction commits, its updates persist permanently in secondary storage and will survive subsequent power failures, crashes, or OS reboots via Write-Ahead Logging (WAL) and checkpointing.

Q10 (15M) **What is Normalization? Explain 1NF, 2NF, 3NF, and BCNF with suitable relational examples and anomaly reduction.**

Definition: Normalization is a formal process of decomposing relations with anomalies into structured relations to eliminate insertion, deletion, and update anomalies while avoiding redundant data storage.

1. First Normal Form (1NF):

- A relation is in 1NF if and only if all attribute values are atomic (no multivalued, composite, or nested attributes).
- *Violation:* `Employee(EmpID, Skills)` where `Skills = "Java, Python, C++"`.
- *Decomposition:* Split into atomic rows: `(101, 'Java')`, `(101, 'Python')`, `(101, 'C++')`.

2. Second Normal Form (2NF):

- Relation must be in 1NF and have **no partial dependency** (no non-prime attribute should depend on a proper subset of any candidate key).
- *Example:* `Enrollment(StudentID, CourseID, CourseFee, Grade)`. Here `CourseFee` depends only on `CourseID` (partial).
- *Decomposition:* `StudentCourses(StudentID, CourseID, Grade)` and `CourseDetails(CourseID, CourseFee)`.

3. Third Normal Form (3NF):

- Relation must be in 2NF and have **no transitive dependency** (for every non-trivial FD X \to Y, either X is a super key or Y is a prime attribute).
- *Example:* `Employee(EmpID, EmpName, DeptNo, DeptHead)`. Here `EmpID → DeptNo` and `DeptNo → DeptHead`, creating transitive dependency.
- *Decomposition:* `Emp(EmpID, EmpName, DeptNo)` and `Dept(DeptNo, DeptHead)`.

4. Boyce-Codd Normal Form (BCNF):

- A stricter version of 3NF where for every non-trivial FD X \to Y, determinant X **must strictly be a Super Key**.

Q11 (15M) Explain SQL Aggregate Functions (COUNT, SUM, AVG, MIN, MAX) with GROUP BY and HAVING clauses. Provide full query demonstrations.

Aggregate Functions: Functions that perform calculations on a set of column values and return a single scalar summary value.

- `COUNT(column)` : Returns total count of non-null values. `COUNT(*)` counts all rows.
- `SUM(column)` : Calculates total numeric sum of values.
- `AVG(column)` : Calculates arithmetic mean of non-null values.
- `MIN(column)` / `MAX(column)` : Returns the minimum / maximum value.

Role of GROUP BY and HAVING:

- `GROUP BY` divides records into distinct groups based on identical column values.
- `HAVING` filters grouped rows after aggregation (whereas `WHERE` filters individual tuples prior to grouping).

Comprehensive SQL Example:

```
-- Schema: Employee (EmpID, EmpName, Salary, DeptID, City)
-- Query: Find DeptID, total employees, average salary, and max salary
-- for departments having more than 3 employees and average salary > 45000:

SELECT
    DeptID,
    COUNT(EmpID) AS TotalEmployees,
    AVG(Salary) AS AverageSalary,
    MAX(Salary) AS HighestSalary,
    MIN(Salary) AS LowestSalary
FROM Employee
WHERE City IN ('Delhi', 'Noida', 'Gurugram')
GROUP BY DeptID
HAVING COUNT(EmpID) > 3 AND AVG(Salary) > 45000
ORDER BY AverageSalary DESC;
```

Q12 (15M) **What is Relational Algebra? Discuss its fundamental operations (Select, Project, Union, Set Difference, Cartesian Product, Rename) and derived Join operation.**

Definition: Relational Algebra is a formal procedural query language that takes one or two relations as input and produces a new relation as output without modifying the underlying stored tables.

Fundamental Operators:

1. **Select (σ):** Unary operator that filters rows satisfying a predicate condition P.

```
σ_{Salary > 50000 AND Dept = 'IT'} (Employee)
```

2. **Project (π):** Unary operator that selects specified columns and eliminates duplicate tuples.

```
π_{EmpName, Salary} (Employee)
```

3. **Union (∪):** Binary operator combining tuples from two union-compatible relations.

```
Student_CS ∪ Student_IT
```

4. **Set Difference (−):** Returns tuples present in relation R but not in S (union compatible).

```
All_Students - Placed_Students
```

5. **Cartesian Product (×):** Cross product pairing every tuple of R with every tuple of S.

```
Employee × Department
```

6. **Rename (ρ):** Renames relation or attributes: \rho_{S(A_1, A_2)} (R).

Derived Operation — Natural Join (⋈): Combines tuples of two relations matching on identical common attribute names:

```
Employee ⋈ Department = π_{...} (σ_{Employee.DeptID = Department.DeptID} (Employee × Department))
```

Q13 (15M) **Draw an ER Diagram for a Hospital Management System specifying entities, attributes, primary keys, relationships, and cardinalities. Show conversion to Relational Schema.**

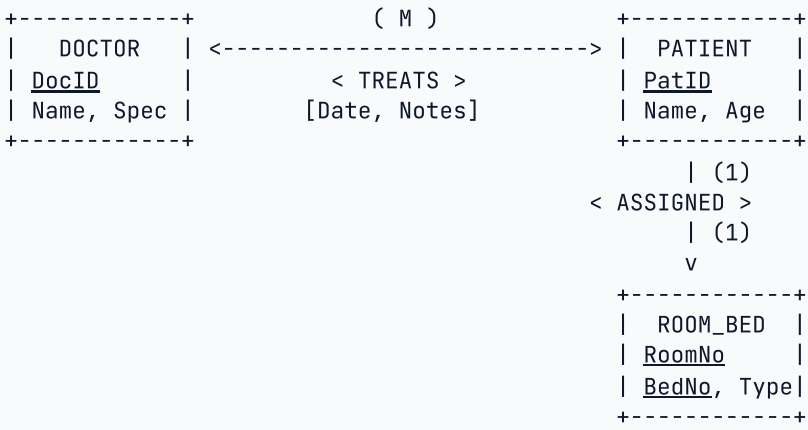
1. Entity Sets & Attributes:

- **DOCTOR**: DocID, Name, Specialty, Phone
- **PATIENT**: PatID, Name, Age, Gender, Address
- **ROOM / BED**: RoomNo, BedNo, RoomType, DailyCharge
- **BILL / ACCOUNT**: BillID, BillDate, TotalAmount, PaymentStatus

2. Relationships & Cardinalities:

- `TREATS` : Many-to-Many between DOCTOR and PATIENT with attribute `DateOfTreatment` .
- `ASSIGNED_TO` : Many-to-One between PATIENT and BED (A patient is assigned to one bed; a bed holds one patient at a time).
- `GENERATES` : One-to-One between PATIENT and BILL.

3. ASCII ER Diagram Representation:



4. Relational Schema Reduction:

1. Doctor(DocID, Name, Specialty, Phone)
2. Patient(PatID, Name, Age, Gender, Address, RoomNo, BedNo)
   FOREIGN KEY (RoomNo, BedNo) REFERENCES Room_Bed(RoomNo, BedNo)
3. Room_Bed(RoomNo, BedNo, RoomType, DailyCharge)
4. Treats(DocID, PatID, TreatmentDate, Prescription)
   FOREIGN KEY (DocID) REFERENCES Doctor(DocID),
   FOREIGN KEY (PatID) REFERENCES Patient(PatID)
5. Bill(BillID, PatID, BillDate, TotalAmount, Status)
   FOREIGN KEY (PatID) REFERENCES Patient(PatID)